

Are any of you passionate about the impact of generative AI on the environment?

Posted by u/Toddeyy • 0 points • 30 comments

Hello! I'm a senior in high school that is currently working on a project highlighting the environmental impact of generative AI. One of the things that we need in this project is perspectives from different individuals that are passionate about this topic, so what are your opinions on generative AI and the impact that it has on our planet?

Comments

u/Silent_Plankton8486 • 2 points • 2026-03-23 08:44 UTC

i'm worried about AI training energy, anyone studying solutions?

u/AfraidKaleidoscope30 • 1 points • 2026-03-22 06:56 UTC

Current impact on environment? Hard to tell. I haven't seen much actual data on it. From my understanding most data centers in U.S. are limited to how much water they can use by local government.

u/jarkon-anderslammer • 1 points • 2026-03-22 03:10 UTC

So I'll preface this by saying I have a bit of a weird perspective here; I spent 10 years as an environmental engineer (PE licensed) before switching to software dev at an AI startup. I get the concern, but I think the "data centers are destroying the planet" take is missing some important context.

The IEA's own numbers show that if AI gets widely adopted in just a few key sectors: grid optimization, building HVAC, transportation routing, industrial processes; the emissions reductions could be 3-4x larger than the total emissions from all data centers combined by 2035. We're talking about stuff like cutting building energy use by 10% through smarter heating/cooling, or shaving 5-10% off transportation fuel use through better routing. Those aren't hypothetical either. Alaska Airlines already saved 1.2 million gallons of fuel in a single year using AI route optimization. Google's claiming 26 million metric tons of avoided emissions in 2024 from just five products. And this exact panic happened before; in the early 2000s everyone was convinced the internet boom would be an energy catastrophe, and then data center energy use basically flatlined from 2005-2017 despite workloads exploding, because efficiency improvements kept pace.

That said; I'm not a pollyanna about this. The real risk is Jevons paradox. As AI gets cheaper and more efficient, it gets embedded in everything, and the sheer volume of use could overwhelm per-unit efficiency gains. Whether this plays out well depends a lot on whether we're intentional about where we point AI, and whether clean energy buildout keeps up. But "data centers bad" is way too simple a take when the net math might actually be strongly positive.

u/Toddeyy • 2 points • 2026-03-22 14:40 UTC

All perspectives are necessary to create a well rounded conversation, so thank you!

u/biscuit_consumer • 1 points • 2026-03-22 03:05 UTC

AI + the enshitification of the internet (and frankly everything) is imo doing the most harm. Modern and old processes that worked perfectly fine and efficient without ai, are being made fully dependent on it.

I bet we're even going to get enshitification of AI because it using more processing power and RAM inflates the AI bubble

u/Toddeyy • 1 points • 2026-03-22 15:01 UTC

I definitely think that it would be better if the use of AI didn't completely overshadow human work, as one of the other comments said, we need a balance

u/twinnedcalcite • 3 points • 2026-03-21 22:18 UTC

I'm going to focus on things from the infrastructure side. I work more in development and infrastructure.

AI is here to stay but I don't think it justifies building so many data centers that fast. I remember the dot com boom and bust (look into this and what resulted from it). This is giving off those vibes. Poorly planned data centers means if the company goes out then there is a piece of real estate that is very hard to convert to something else.

There is a very interesting [XKCD WhatIf page dealing with google data centers](https://what-if.xkcd.com/63/). Alphabet (google's parent company) has been using AI in the background for ages. What the public is seeing a something they can access vs being hidden away in the development side of things. They are not asking for more data centers. Neither is microsoft. So look into which companies want these data centers and whose behind them. Follow the money.

Now onto the data centers themselves.

They need to be buildings with a very strong foundation, servers are heavy and cooling units are heavy. They need minimal super vision so only a small physical office is needed.

They are very energy intensive so need special power lines for supply. Those power lines will feed of the grid fully unless they install a geothermal/water cooled system, solar panels, or something else. They also need MASSIVE back up generators. back up generators are almost always diesel. They are robust and reliable with fairly simple maintenance.

So the building needs to be approved by the local government AND the utilities before it ever breaks ground. This takes time and should not be rushed as this data center has a live span of decades. If the local area cannot support it either by their water usage or power generation capabilities, it should not be approved.

That approval is where the fear lies. The US federal government has removed and is bulling areas to build these data centers without the proper checks and balances. Which means there is no incentive for them to be efficient and to coordinate with the local utility. We've seen it with detention centers.

Piss pour planning gets people killed. Regulations were written in blood. Don't ignore them. If it's done right there will be little press as they don't like to cover good negotiations unless it's going above and beyond the standards.

u/Toddeyy • 1 points • 2026-03-21 22:30 UTC

This is also a really important aspect that I haven't thought too much into as well, which makes it all the more important to bring light to. I will definitely make an effort to pay more attention to this, thank you

u/DontbegayinIndiana • 1 points • 2026-03-21 21:23 UTC

I really appreciated this fact-check-y video by Hank Green: [https://youtu.be/H_c6MWk7PQc?si=HzNUDwVNf70C2IXF](https://youtu.be/H_c6MWk7PQc?si=HzNUDwVNf70C2IXF)

My concern is that we're focused so much on this new technology that genuinely could help us make things more efficient (although I don't know if the AI I'm thinking of here is considered "generative" as it's more just analytical, but Mustafa Suleyman talks about it in "What If We Get It Right?" by Ayana Elizabeth Johnson), when we still have things that are generally pretty useless taking up resources (*cough* lawns *cough*). I understand we need to tackle this from every angle, but I feel AI is going to get a lot of resistance as it is new technology (maybe it'll be easier since it's not part of everyday yet though) when the lawns of some data centers take more water than the actual data processing does.

That said, many uses of AI currently are not helpful and take meaningful work from artists. I just think we shouldn't write AI off entirely, because (and maybe someone can explain this to me) isn't all computing technically AI?

u/SallyStranger • 0 points • 2026-03-21 22:49 UTC

That many other things are wasteful and/or useless is not an argument for tolerating data centers which are definitely wasteful and mostly useless. That this fallacious argument gets trotted out so often in their favor suggests that there's a lack of any other better arguments for accepting them.

u/Toddeyy • 1 points • 2026-03-21 22:32 UTC

Completely agree, I can see how it can have benefits within computer programming and security protocols, and I also agree that we focus too much on the fact that it's new instead of the genuine effects that it has on the environment, thanks for proving your perspective on this

u/Toddeyy • 1 points • 2026-03-21 22:33 UTC

*providing, sorry lol